PLAN-04 - Scope, Requirements, and Business Rules

POS-DGFY - Global Location Switching | Plan v1.0 | Revision 2026-09-27

Field	Value
Project	POS-DGFY - Global Location Switching
Plan version	1.0
Section	PLAN-04 - Scope, Requirements, and Business Rules
Revision date	2026-09-27
Plan status	NOT EXECUTION READY
Purpose	Define traceable functional, security, data, operational, and migration requirements.
Dependencies	PLAN-01 through PLAN-03; Phase 354 scope matrix.

In scope

- Global location entry point and context labels.
- Operating-location switching through the existing Shift workflow.
- Location-aware Items, stock, Add Item, sales, parked sales, queue, reports, shifts, settings, vouchers, price lists, affiliates, and audit.
- Server-side permissions, concurrency, idempotency, offline safety, testing, rollout, observability, and recovery.

Out of scope

- GPS, Wi-Fi, IP, or automatic physical branch detection.
- Automatic conflict takeover or silent shift reassignment.
- Duplicate branch catalogs or destructive schema replacement.
- New offline login or branch transfer behavior not already supported by the contract.

Functional requirements

ID	Requirement	Priority	Verification
FR-001	Authorized users can open a global Current Location entry point.	P1	Desktop/mobile UI test
FR-002	The UI distinguishes operating, viewing, and queue context.	P1	UI and E2E test
FR-003	An active operating switch uses the governed close/open shift flow.	P0	API integration test
FR-004	Only authorized destination locations and terminals are selectable.	P0	Authorization test
FR-005	Sales, stock, queue, parked sales, and payments use the operating location.	P0	Cross-module E2E
FR-006	Reports and audit support authorized viewing-location filters.	P1	Report/API test
FR-007	Settings, vouchers, price lists, and affiliates apply explicit scope rules.	P1	Module contract tests
FR-008	Every successful or rejected transition is traceable.	P0	Audit verification

Quality, security, data, and operations requirements

ID	Requirement	Priority
NFR-001	No stale request from the old context may overwrite the new context.	P0
NFR-002	Switch submissions are idempotent and safe to retry.	P0
SEC-001	Backend rejects forged tenant, location, terminal, shift, cashier, and operator identifiers.	P0
SEC-002	Location permission is independent from action permission.	P0
DATA-001	Every completed sale has exactly one immutable location snapshot.	P0
DATA-002	Stock truth remains location-scoped and reconciliable.	P0
OPS-001	Transition, denial, conflict, and reconciliation signals are observable.	P1
MIG-001	Schema changes are additive and backward compatible until verified.	P0

Business rules

ID	Rule	Failure behavior	Affected areas
BR-001	A sale belongs to one operating location and active shift.	Reject or preserve current state.	Sales, inventory, reports, audit
BR-002	Switch permission does not grant destination access.	Return 403.	Authorization, UI
BR-003	No switch with unresolved cart, payment, or offline operation.	Block with actionable reason.	Checkout, shift, offline
BR-004	Source shift close and destination shift open are atomic.	Keep source state; retry safely.	Shift, cash, audit
BR-005	Viewing a location never grants mutation or incoming-queue access.	Return 403 or read-only state.	Reports, monitor, queue
BR-006	Parked sales retain origin location and shift ownership.	Require explicit handoff policy.	Parked sales, payments
BR-007	Historical transactions do not change when current context changes.	Reconcile and block release if violated.	Reports, finance
BR-008	Strict terminal mode requires a destination registered terminal.	Reject destination.	Terminal, shift

Acceptance criteria

- AC-001: A permitted user can switch safely and the new location appears consistently in all affected modules.
- AC-002: An unassigned location, forged location ID, or missing permission is rejected server-side.
- AC-003: Pending offline work, unresolved cart, or concurrent transition cannot corrupt the shift.
- AC-004: Stock, queue, sales, reports, vouchers, pricing, affiliates, and audit use the correct context.
- AC-005: Desktop and mobile users can identify current context and understand blocked actions.
- AC-006: Rollback preserves historical transactions and supports reconciliation.